

ECON 5345 Homework 8 Report

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April 10, 2026

Setup: The central planner maximizes the following objective.

$$\max_{C_t, K_t, L_t, I_t} \mathbb{E} \sum_{t=0}^{+\infty} \beta^t \left\{ \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\xi L_t^{1+1/\eta}}{1+1/\eta} \right\},$$

subject to

$$\begin{aligned} Y_t &= Z_t K_{t-1}^\alpha L_t^{1-\alpha}, \\ Y_t &= C_t + I_t + \frac{\phi}{2} \left(\frac{K_t}{K_{t-1}} - 1 \right)^2 K_{t-1}, \\ K_t &= (1 - \delta) K_{t-1} + I_t, \\ Z_t &= e^{\varepsilon_t} Z_{t-1}^\rho, \varepsilon_t \stackrel{i.i.d.}{\sim} (0, \sigma_\varepsilon^2). \end{aligned}$$

Part I

(a) Combining the first three constraints, we have

$$Z_t K_{t-1}^\alpha L_t^{1-\alpha} = C_t + [K_t - (1 - \delta)K_{t-1}] + \frac{\phi}{2} \left(\frac{K_t}{K_{t-1}} - 1 \right)^2 K_{t-1}. \quad (1)$$

Then the maximization problem is

$$\max_{C_t, K_t, L_t} \mathbb{E} \sum_{t=0}^{+\infty} \beta^t \left\{ \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\xi L_t^{1+1/\eta}}{1+1/\eta} \right\}, \quad (2)$$

subject to constraint (1).

The Lagrangian is

$$\begin{aligned} \mathcal{L} = \mathbb{E} \sum_{t=0}^{+\infty} \beta^t & \left\{ \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\xi L_t^{1+1/\eta}}{1+1/\eta} \right] \right. \\ & \left. + \lambda_t \left[Z_t K_{t-1}^\alpha L_t^{1-\alpha} - C_t - [K_t - (1 - \delta)K_{t-1}] - \frac{\phi}{2} \left(\frac{K_t}{K_{t-1}} - 1 \right)^2 K_{t-1} \right] \right\}, \end{aligned} \quad (3)$$

where λ_t is the attached Lagrange multiplier. Taking FOCs with respect to C_t , K_t , and L_t , we get

$$[C_t] : \quad C_t^{-\sigma} = \lambda_t, \quad (4)$$

$$\begin{aligned} [K_t] : \quad & -\lambda_t \left[\phi \left(\frac{K_t}{K_{t-1}} + 1 \right) - 1 \right] \\ & + \beta \mathbb{E}_t \lambda_{t+1} \left\{ \alpha \frac{Y_{t+1}}{K_t} + (1 - \delta) - \frac{\phi}{2} \left[1 - \left(\frac{K_{t+1}}{K_t} \right)^2 \right] \right\} = 0, \end{aligned} \quad (5)$$

$$[L_t] : \quad -\xi L_t^{1/\eta} + \lambda_t (1 - \alpha) \frac{Y_t}{L_t} = 0. \quad (6)$$

- (b) Without stochasticity, we normalize $Z = 1$. Denote X^* as the steady state value of a variable X . We need to solve for $(C^*, K^*, L^*, I^*, Y^*)$, and the prices (R^*, W^*) where R^* is the capital rental rate and W^* is the wage rate. Substituting (4) into (5) and (6), together with the constraints and the expressions for capital rental rate and wage rate, we get the system:

$$\frac{1}{\beta} = R^* + (1 - \delta), \quad (7)$$

$$W^* = \xi(L^*)^{1/\eta}(C^*)^\sigma, \quad (8)$$

$$Y^* = (K^*)^\alpha(L^*)^{1-\alpha}, \quad (9)$$

$$Y^* = C^* + I^*, \quad (10)$$

$$I^* = \delta K^*, \quad (11)$$

$$R^* = \alpha \frac{Y^*}{K^*}, \quad (12)$$

$$W^* = (1 - \alpha) \frac{Y^*}{L^*}, \quad (13)$$

where (10) is derived by plugging the LoM. With 7 variables and 7 equations, the system is just-solvable.

- (c) Denote $\hat{x}_t = \frac{X_t - X^*}{X^*}$. Combining (4) and (5), we can log linearize the derived FOC of K_t to get

$$\sigma(\mathbb{E}_t \hat{c}_{t+1} - \hat{c}_t) = \beta R^* \mathbb{E}_t(\hat{y}_{t+1} - \hat{k}_t) + \beta \phi \mathbb{E}_t(\hat{k}_{t+1} - \hat{k}_t) - \phi(\hat{k}_t - \hat{k}_{t-1}). \quad (14)$$

where (7) is used to substitute $R^* + (1 - \delta) = 1/\beta$.

Combining (4) and (6), we can log linearize the derived FOC of L_t to get

$$(1 + 1/\eta)\hat{\ell}_t = -\sigma\hat{c}_t + \hat{y}_t. \quad (15)$$

The production function can be log linearized as

$$\hat{y}_t = \hat{z}_t + \alpha\hat{k}_{t-1} + (1 - \alpha)\hat{\ell}_t. \quad (16)$$

The consumption-investment equation can be log linearized as

$$\hat{y}_t = \hat{c}_t \frac{C^*}{Y^*} + \hat{i}_t \frac{I^*}{Y^*}. \quad (17)$$

The LoM can be log linearized as

$$\hat{k}_t = (1 - \delta)\hat{k}_{t-1} + \delta\hat{i}_t. \quad (18)$$

The prices can be log linearized as

$$\hat{r}_t = \hat{y}_t - \hat{k}_{t-1}, \quad (19)$$

$$\hat{w}_t = \hat{y}_t - \hat{\ell}_t. \quad (20)$$

The productivity can be log linearized as

$$\hat{z}_t = \rho\hat{z}_{t-1} + \varepsilon_t \quad (21)$$

- (d) α, δ, ρ , and σ_ε can be estimated by OLS. η and σ are jointly determined by (15). With a steady state R^* , we can use the estimated δ to derive the beta, and ϕ can be estimated by (14) after getting the estimated σ and derived β . ξ cannot be estimated.

Part II

(e) By (7), we have

$$R^* = \frac{1}{\beta} - (1 - \delta) = 0.0351.$$

This 3% interest rate is reasonable. By (12), we have

$$\frac{K^*}{Y^*} = \frac{\alpha}{R^*} = 9.4967.$$

This is also reasonable. By (11) and (10), we have

$$\frac{I^*}{Y^*} = \delta \frac{K^*}{Y^*} = 0.2374, \quad \text{and} \quad \frac{C^*}{Y^*} = 1 - \frac{I^*}{Y^*} = 0.7626.$$

Both are reasonable. By (9), we have

$$\frac{L^*}{Y^*} = \left(\frac{K^*}{Y^*} \right)^{-\frac{\alpha}{1-\alpha}} = 0.3245$$

Combining (8) with (13), we have

$$(1 - \alpha) \frac{Y^*}{L^*} = \xi (L^*)^{1/\eta} (C^*)^\sigma.$$

Substituting the parameter values, we get

$$L^* = \sqrt{\frac{2}{3} \left(\frac{C^*}{Y^*} \right)^{-1}} = 0.9350.$$

This seems not reasonable because the steady state labour hours should not consist of more than 90% of the time. The problem may be because of ξ being too small.

For ϕ , (14) implies that capital stock and consumption data will be used, together with forward expectations of output and capital stock. Relevant data can be the stock prices which provides some information on expected output and capital stock.

(f) Figure 1 shows the IRFs.

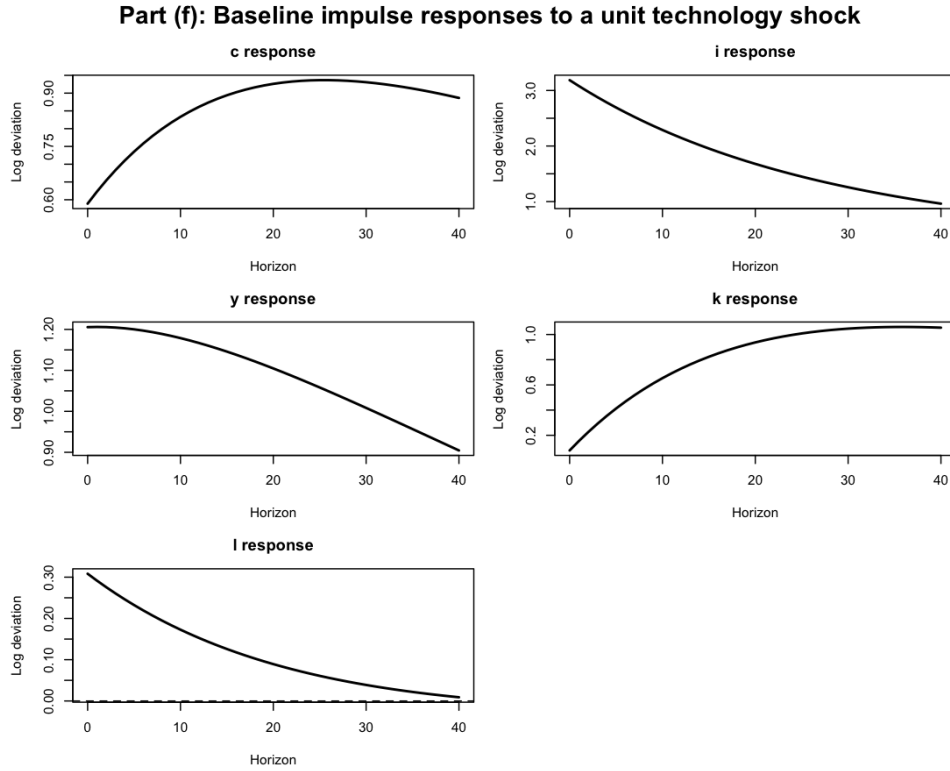


Figure 1: IRFs to a unit technology shock

Following a positive technology shock, all variables increase on impact. Consumption and capital rise gradually and converge to higher steady-state levels. Investment is initially high but declines over time as capital accumulates. Output rises on impact but decreases thereafter toward its long-run level. Labor declines monotonically due to the wealth effect and converges back to steady state.

(g) We show the comparison parameter by parameter.

Figure (2) shows the comparison of η .

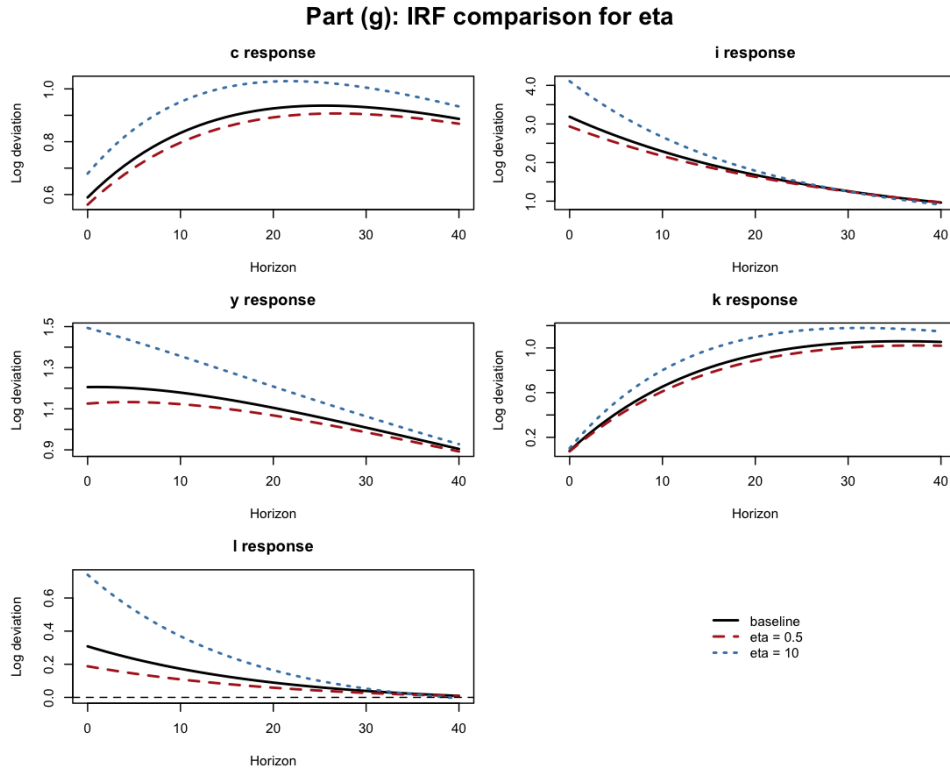


Figure 2: Comparison of η

η is the Frisch elasticity of labour supply. A higher η means that workers are more responsive to changes in real wages. After a positive productivity shock, real wage increases so that labour hours increase more when η is higher. As a result, output increases more, consumption increases more, investment increases more right after the shock. As a result, capital accumulation is faster.

Figure (3) shows the comparison of ϕ .

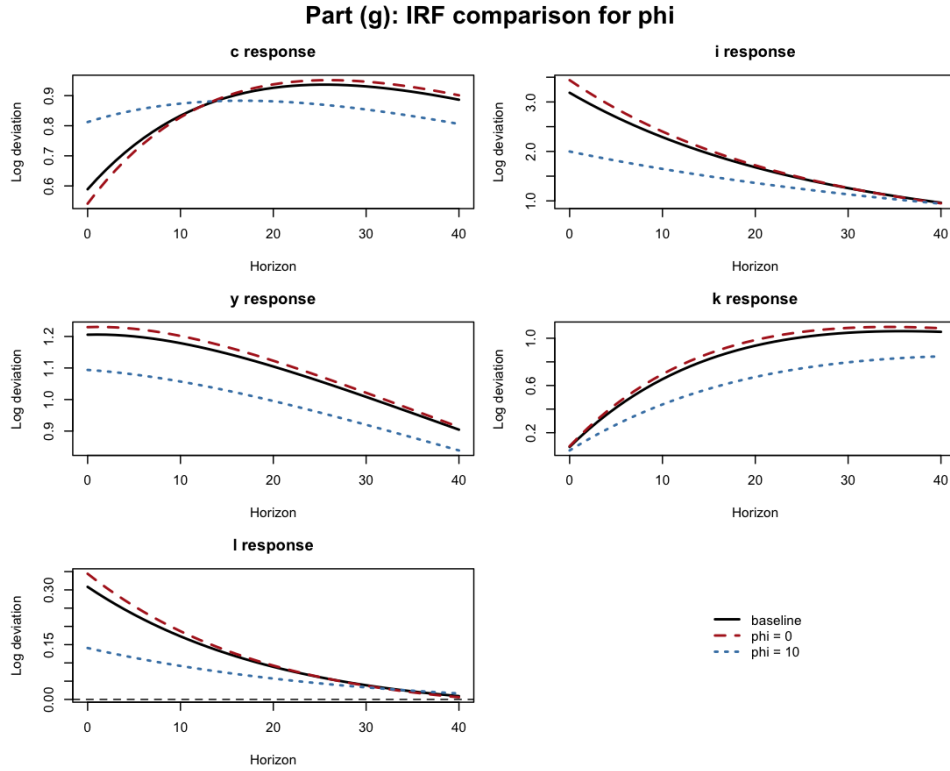


Figure 3: Comparison of ϕ

ϕ is the adjustment cost parameter. A higher ϕ means that it is more costly to adjust capital stock. This limits the extent to which output and investment increase. Since investment cannot respond as much, consumption rises more than the baseline. Labour increases less than the baseline due to the smaller response of real wage. In the long-run, capital stock converges to a lower steady-state due to the higher adjustment cost, and so do output and consumption.

Figure (4) shows the comparison of σ .

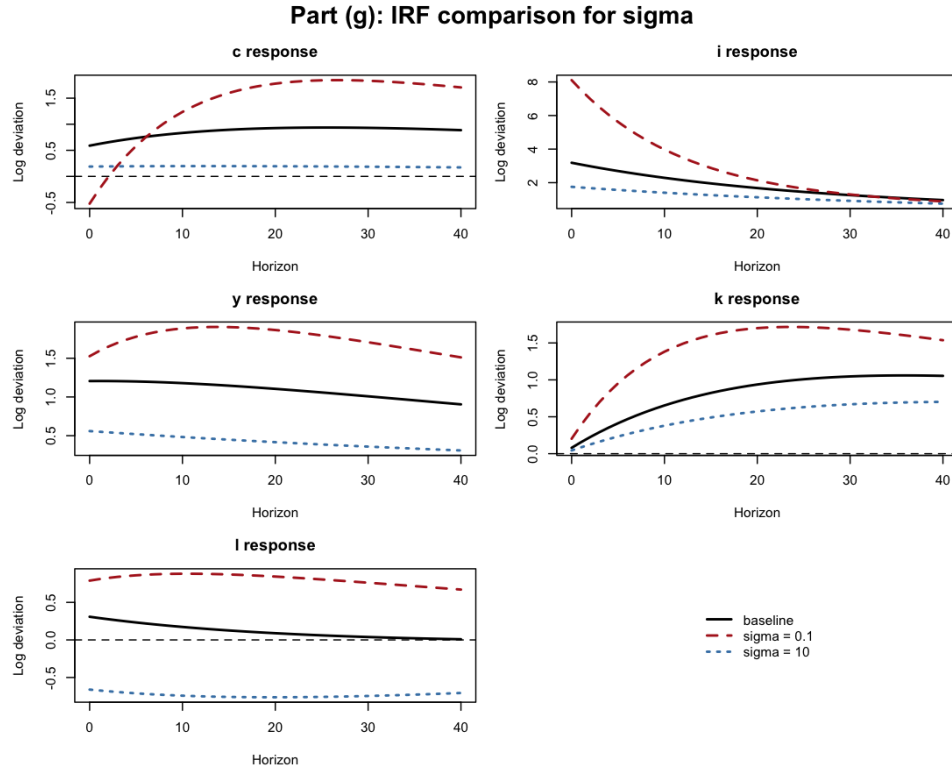


Figure 4: Comparison of σ

$1/\sigma$ is the intertemporal elasticity of substitution. When $\sigma = 10$, the IES is small, meaning that consumers are reluctant to change their consumption. As a result, consumption increases less and output increases less. As the consumption smoothing effect is strong, a higher real wage leads to less labour supply, so labour drops after the shock. Investment also increases less due to the consumption smoothing incentive. When $\sigma = 0.1$, the IES is large, meaning that consumers are willing to consume in the future. As a result, consumption drops immediately after the shock but rises to a higher value in the long run. This incentive also leads to an increase in labour to gain more income, and higher saving (investment), and thus higher output and capital stock.

Figure (5) shows the comparison of ρ .

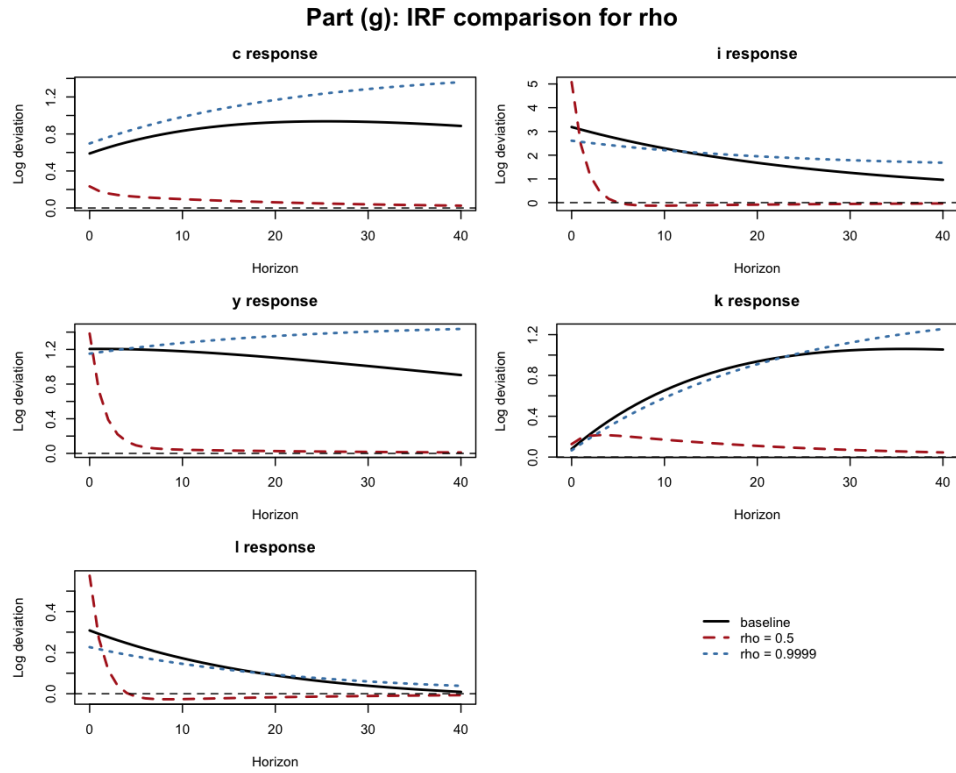


Figure 5: Comparison of ρ

ρ is the persistence of the technology shock. A higher ρ means that the technology shock is more persistent. This is straightforward to see from the IRFs.

- (h) Table 1 summarizes the variance-covariance matrix.

Table 1: Variance-Covariance Matrix of Variables

	dc	dy	$d\ell$	di
dc	1.7295	3.4489	0.8597	8.9721
dy	3.4489	7.0667	1.8089	18.6871
$d\ell$	0.8597	1.8089	0.4746	4.8575
di	8.9721	18.6871	4.8575	49.8930

Note: All entries are expressed in units of 10^{-5} .

For the data counterpart, I use real gross domestic product, real personal consumption expenditures, average weekly hours of all employees, and gross private domestic investment from FRED. Table 2 shows the variance-covariance matrix of the HP-filtered data counterparts. VAR is not used for this question.

Table 2: Variance-Covariance Matrix of Variables (HP-Filtered Data Analogues)

	dC	dY	dL	dI
dC	24.3422	19.7796	1.6201	57.2742
dY	19.7796	18.9462	1.8634	65.7208
dL	1.6201	1.8634	1.7509	14.9673
dI	57.2742	65.7208	14.9673	382.6942

Note: All entries are expressed in units of 10^{-5} using HP-filtered log data with quarterly smoothing parameter $\lambda = 1600$.

The magnitudes differ a lot. The main reason may be that the model only considers the TFP shock while the data counterparts experience much more shocks than the TFP shock. There is also the magnitude issue, where the aggregate data seems much larger than the normalized model.

- (i) Main variables respond in a similar pattern except labour. My results show that labour hour increases while Gali's VAR result shows that labour hour decreases. The main difference is that Gali's model incorporates money. In a monetary model, when labour is introduced, money neutrality does not hold and money growth affects labour supply decision. In this model, the mechanism is such that a positive productivity growth increases the real wage which leads to higher labour supply. In Gali's model, when the coefficient of lagged TFP shock is positive in the money growth process, a positive productivity growth leads to higher expected inflation and thus a higher nominal interest rate. As a result, people hold less money and thus less consumption. As a result, labour hours decrease.

Part III